

BOLTENHAGEN, Germany

WMO: 101610

Lat: **54.0028N**Lon: **11.1908E**Elev: **52**StdP: **14.67**Time Zone: **1.00 (EUC)**Period: **82-03**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.7	18.8	8.3	8.4	16.3	12.3	10.3	19.2	36.9	38.9	33.7	37.9	8.8	220	0.689

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	77.2	66.6	73.7	64.8	71.0	63.4	67.7	74.7	65.9	72.3	64.4	69.9	10.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.8	92.0	71.7	63.3	87.3	69.9	61.8	82.8	68.1	32.1	74.8	30.7	72.5	29.5	70.1	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.7	28.0	25.0	13.1	86.2	9.7	5.3	6.1	90.0	0.4	93.1	-5.0	96.1	-12.1	100.0		
(5)			12.3	72.0	9.5	1.9	5.5	73.3	0.0	74.4	-5.4	75.5	-12.2	76.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.9	34.3	33.7	39.3	44.5	52.5	58.5	62.9	62.6	56.9	50.1	41.5	37.2
	DBStd	11.91	8.84	7.67	6.34	5.49	5.31	5.13	4.62	4.66	4.16	5.64	5.55	6.20
	HDD50	2214	485	456	335	181	33	1	0	0	1	66	258	398
	HDD65	6309	950	875	798	615	388	205	97	105	248	461	705	862
	CDD50	1456	0	1	2	16	111	256	399	390	207	70	3	1
	CDD65	76	0	0	0	0	1	9	32	30	4	0	0	0
	CDH74	301	0	0	0	0	4	28	121	133	15	0	0	0
	CDH80	57	0	0	0	0	0	3	21	31	2	0	0	0
	WSAvg	12.3	15.1	14.1	12.9	11.4	10.9	10.3	10.7	10.3	11.6	12.4	13.5	14.1
	PrecAvg	24.9	2.0	1.6	1.5	1.3	2.0	2.5	2.5	3.0	2.2	2.0	2.0	2.2
(15) Precipitation	PrecMax	32.6	4.1	3.3	4.2	2.7	4.3	5.9	5.8	7.6	6.2	4.6	4.9	5.1
	PrecMin	16.7	0.0	0.1	0.2	0.0	0.4	0.2	0.3	0.7	0.8	0.7	0.2	0.7
	PrecStd	4.2	1.0	0.8	0.8	0.7	1.0	1.3	1.4	1.7	1.3	0.8	1.2	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.3	56.6	59.5	67.9	73.5	77.7	82.7	83.3	75.3	69.2	56.2	53.5
		MCWB	49.5	52.2	51.8	55.0	62.3	66.7	68.4	67.1	66.0	63.1	52.5	49.9
	2%	DB	48.6	51.1	53.8	60.7	67.1	72.5	77.0	77.3	69.6	63.4	53.2	51.0
		MCWB	46.0	47.5	48.6	51.8	59.4	64.4	67.0	66.3	61.4	58.0	50.4	48.5
	5%	DB	46.9	46.2	50.0	56.3	63.2	69.6	73.6	73.0	66.1	60.0	51.2	48.6
		MCWB	44.6	43.7	46.1	49.9	57.3	62.7	65.3	64.5	59.8	55.7	48.9	46.5
	10%	DB	44.8	43.4	47.5	52.7	60.7	67.1	70.7	70.2	63.5	57.8	49.0	46.1
		MCWB	42.6	41.1	44.3	47.9	55.3	60.8	63.7	63.3	58.0	54.0	47.0	44.4
	0.4%	WB	49.2	52.5	52.7	56.8	62.9	68.0	70.6	70.1	66.0	62.8	53.8	50.7
		MCDB	51.4	56.3	58.5	63.7	71.6	74.9	79.5	78.6	74.2	68.6	55.6	52.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	46.5	47.4	49.3	52.9	60.3	65.6	67.9	67.6	63.1	59.0	50.9	48.7
		MCDB	48.1	50.4	52.9	59.6	65.9	71.2	75.2	74.5	67.7	62.4	52.6	50.7
	5%	WB	44.9	43.9	46.8	50.5	58.1	63.7	66.1	65.7	60.7	56.3	49.0	46.6
		MCDB	46.8	46.1	49.8	55.5	62.6	68.5	72.4	71.9	64.8	59.1	50.9	48.4
	10%	WB	42.8	41.2	44.4	48.3	55.9	61.6	64.4	64.0	58.7	54.6	47.3	44.3
		MCDB	44.7	43.2	47.2	52.1	60.0	66.2	69.8	69.4	62.5	57.1	49.0	45.9
(35) Mean Daily Temperature Range	5% DB	MDBR	7.2	7.7	9.1	10.8	11.0	11.7	12.9	13.0	11.1	9.8	7.6	6.6
		MCDBR	8.3	11.2	14.0	17.7	15.8	15.2	17.4	17.3	15.2	12.8	10.1	8.7
		MCWBR	7.6	9.2	10.2	11.2	10.1	10.0	10.2	9.7	9.9	9.1	8.9	8.1
	5% WB	MCDBR	8.3	11.4	13.7	16.4	14.6	15.9	15.0	13.8	11.7	9.6	8.3	8.3
		MCWBR	7.9	9.6	10.8	11.0	9.9	10.1	10.4	9.3	10.3	9.2	8.7	8.0
(40) Clear-Sky Solar Irradiance	taub		0.319	0.340	0.373	0.410	0.402	0.399	0.419	0.415	0.385	0.360	0.338	0.311
	taud		2.311	2.315	2.255	2.206	2.291	2.323	2.288	2.311	2.378	2.410	2.367	2.307
	Ebn at Noon		200	236	252	258	268	270	261	254	248	227	192	177
	Edh at Noon		19	25	33	40	39	38	39	36	30	23	18	16
(44) All-Sky Solar Radiation	RadAvg		167	374	768	1292	1636	1729	1620	1324	964	516	214	123
	RadStd		23	56	75	167	193	123	209	98	102	65	24	8

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-261	N/A	N/A	

Nomenclature: See separate page